R5.2: how much of a binary isogeny class is within GHS reach?

Round 5, lead R5.2. 2026-10-18. Example `examples/r5_2_ghs_class.rs`. Records: `census.json`, `walk.json`, `walk_20.json`, `selftest.json` (this directory). Toy fields F_{2^N}, $N \in \{12, 14, 15, 16, 18, 20\}$, exhaustive over $b \in K^*$ and both twists. No deployed-curve claim. Prime N is out of scope: GHS needs $N = ln$ with $n \geq 2$.

Verdict

- **Cost model P** (min of Gaudry $g! \cdot q^{2-\frac{2}{g}}$ and Enge–Gaudry $L_{q^g}\left[\frac{1}{2}, \sqrt{2}\right]$, against rho $\sqrt{\pi r/4}$): **0 weak curves** out of 2 858 996 at every N (derived verdicts on measured g, r). The best class is 1.65–2.42 bits above rho. The hypothesis, read as a cost statement, is **inconclusive**. At these sizes it is refuted only trivially, because no curve is weak at all.
- **Structural low-genus reach** (LG: applicable descent, transfer possible, $g \leq 2n$, $n \geq 5$): a small nonzero share, 0.09 %–3.1 % of curves where it exists. Every class containing such a curve is mixed (median in-class share 3.6 %–25 %). With $\ell \in \{3, 5, 7\}$, 66 %–100 % of the strong curves in those classes reach one. The mean distance is 1.3–3.0 steps (max 15), and a search expands on average 1.5–8.4 curves (max 53). **Confirmed structurally**, with 0 %–34 % of strong curves unreachable via $\ell \leq 7$.
- The constant-free exponent model M1 and the fixed-n exponent model A are degenerate. Over F_2 and F_4, $q^{2-\frac{2}{g}} \leq 16$ whatever g is; and every curve with $n \geq 4$ passes A.
- *Derived from recalled exponents:* at fixed n, transfer needs $g \geq n$, so the GHS exponent $2 - \frac{2}{g}$ is never below Gaudry's decomposition exponent $2 - \frac{2}{n}$, which applies to every curve. Exponent separation inside a class needs n to grow.
- *Library:* `ec_trapdoor::audit_curve` disagrees with the derived (m, g) on 1 248 of 1 500 sampled N = 12 rows; the $\sqrt{}$b-orbit rank itself agrees on 1 500/1 500.

Formula implemented

$\sigma(x) = x^{2^l}$, $c_i = \sigma^i\left(\sqrt{b}\right)$, $U = \mathrm{Span}_{F_2}\{(c_i, 1)\}$, $m = \dim U$, $m' = \deg f$ with $f = \mathrm{Ord}_\sigma\left(\sqrt{b}\right)$. Then $m = m' + [f(1) \neq 0]$ (checked on every curve: 0 mismatches). The genus is $g = 2^{m-1} - 1$ if $(x+1) \nmid f$, else 2^{m-1}. This follows from summing the genera $[c_u \neq 0 \wedge d_u \neq 0]$ of the quadratic subfields. Hand-checked cases: $b = 1$ gives $(1, 1)$; $\sqrt{b} = c + \sigma(c)$ at $(l, n) = (4, 3)$ gives $(3, 3)$; generic b gives $(3, 4)$ at $n = 3$ and $(4, 8)$ at $n = 4$. Point counts come from one Walsh–Hadamard transform. Validation: 1 979/1 979 against a naive count and 80/80 against the library. 2-isogenies are Galois conjugations and preserve m (1 000/1 000).

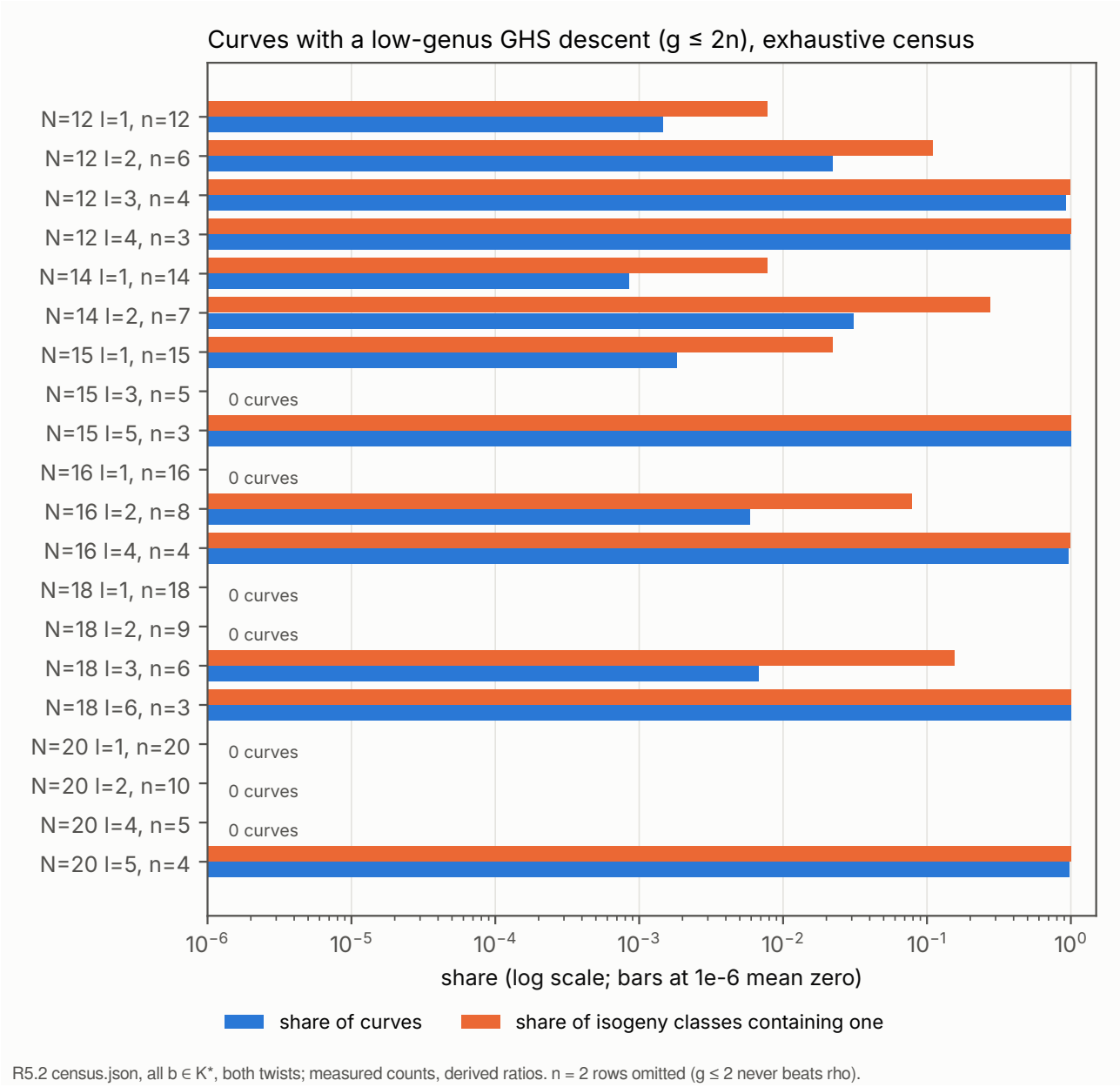


Figure 1: Share of curves (blue) and of isogeny classes (orange) with a low-genus descent, $g \leq 2n$, per factorisation with $n \geq 3$. Measured counts from `census.json`. Rows with $n = 3, 4$ cover almost all curves; rows with $n \geq 5$ are small or zero.

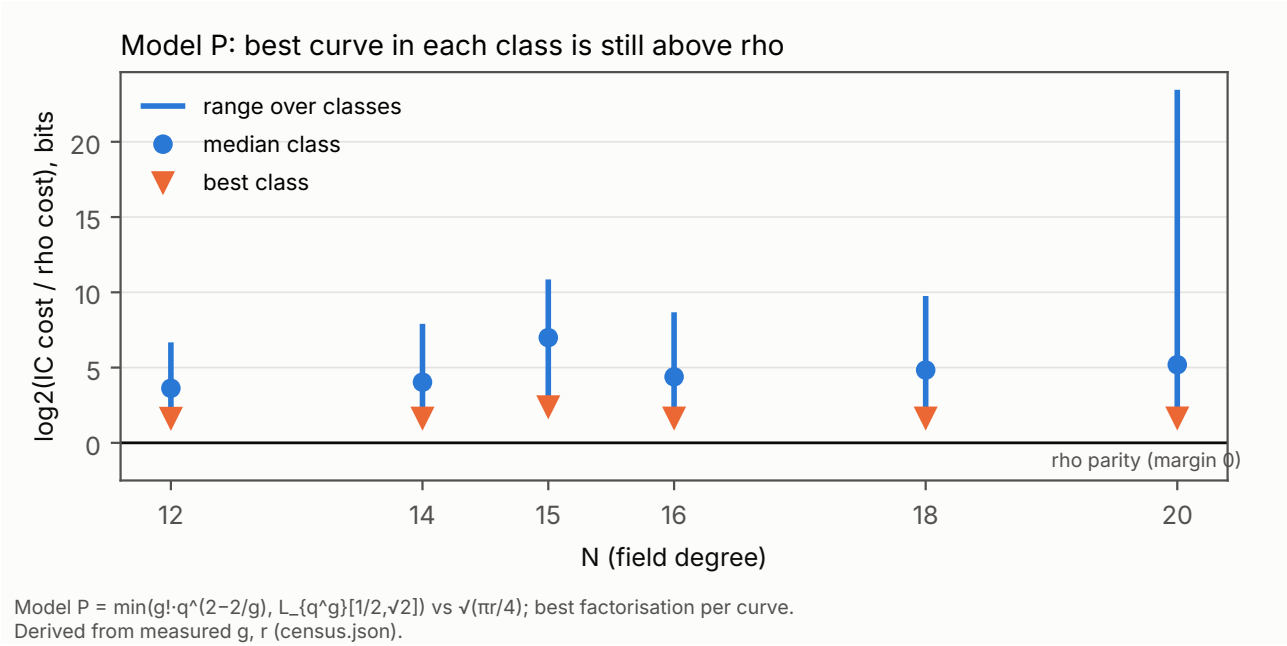


Figure 2: Model P: $\log_2$ of the IC cost over the rho cost, for the best curve in each class (best factorisation per curve). Range, median and minimum over the classes for each N. Derived from measured g and r. No class reaches 0.

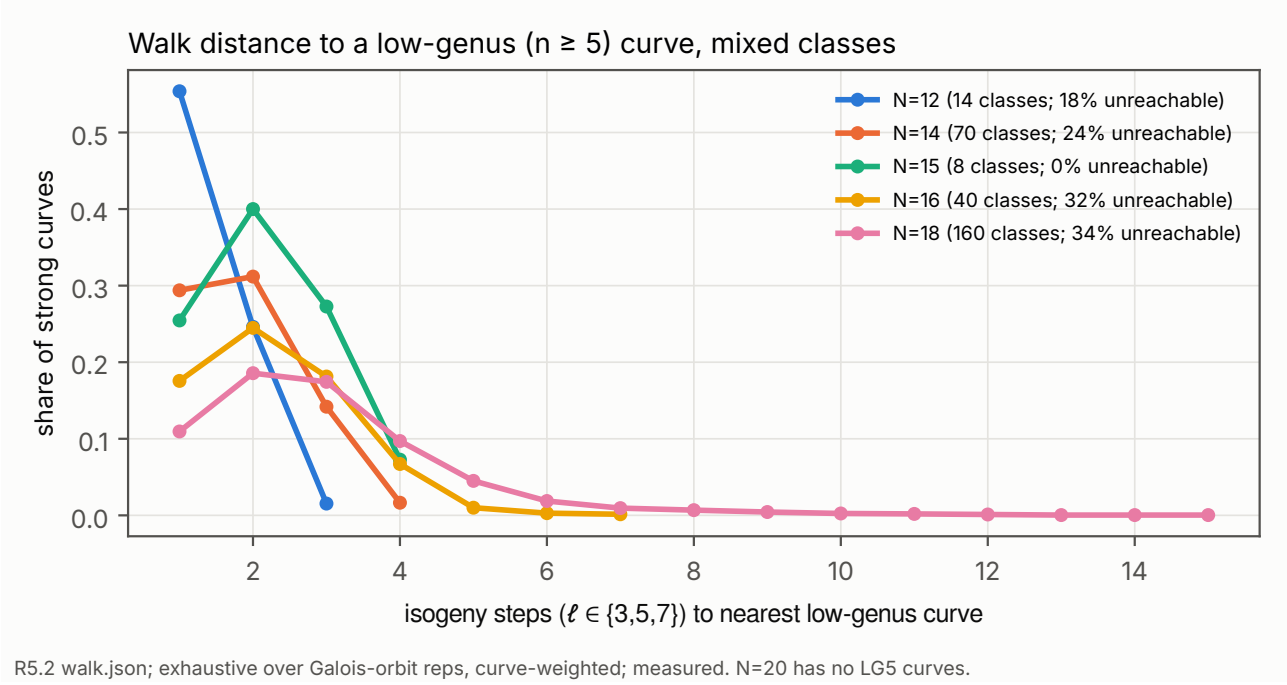


Figure 3: Distance in ℓ-isogeny steps ($\ell \in \{3, 5, 7\}$, Elkies kernels from ψ_ℓ, binary Vélu codomain) from each strong curve to the nearest LG curve with $n \geq 5$, in mixed classes. Curve-weighted and exhaustive, from `walk.json`. Validation: 0 codomain-trace mismatches and 0 asymmetric edges.

Limits

No descent was constructed and no Jacobian order was computed: transfer is the Hasse–Weil size test. The Menezes–Qu applicability condition, the genus-sum formula and the IC constants are recalled. P and LG were defined after the first census showed that M1 and A are degenerate. Only $\ell \leq 7$. Toy $N \leq 20$. Full tables, commands and the library comparison are in `RESULTS.md`.

References (recalled where marked in RESULTS.md)

Gaudry, Hess, Smart, J. Cryptology 15 (2002). Galbraith, Hess, Smart, EUROCRYPT 2002. Menezes, Qu, CT-RSA 2001. Menezes, Teske, Weng, CT-RSA 2004. Gaudry, J. Symb. Comput. 44 (2009). Enge, Gaudry, Acta Arith. 102 (2002).